

# Email Header Analysis by AUannat Uthman

## Summary

A suspected phishing email has been assigned for analysis. The format is .txt file.

It was analyzed with notepad and MXToolbox. One of the servers in the received header is found to be blacklisted. No spf was found and it can be concluded that it was indeed a phishing attempt with a medium risk.

## Headers

```
OriginalChecksum:06D5E44D57FDABA2A9FF26571D32632925A077655904CD0FE295CB7E1AE869F9;UpperCasedChecksum:CFA1D3E3C2E07B798603D0669ED9D0AAC7E2276D64994795B6E0A26A63708FE3;S
izeAsReceived:394;Count:12
From: Microsoft account team ,_<no-reply@access-accsecurity.com>
Subject: Microsoft account unusual signin activity
To: phishing@pot
Content-Length: 22448528
Content-Length: 49413
Date: Wed, 16 Aug 2023 00:15:46 +0000
Reply-To: solutionteamrecognizd03@gmail.com
Content-Type: text/html; charset="UTF-8"
Content-Transfer-Encoding: 8bit
X-IncomingHeaderCount: 12
Message-ID:
<06550bd4-ce4e-4e8c-acf9-f8936085be09@M2NAM04FT048.eop-NAM04.prod.protection.outlook.com>
Return-Path: bounce@atujpdfgher.co.uk
X-MS-Exchange-Organization-ExpirationStartTime: 16 Aug 2023 00:15:46.8420
```

**From** – The sender's email address and name.

**Reply-To** – The visible address to send a reply to.

**To** – The recipient of the email.

**Date** – The date and time the email was sent.

**Subject** – The title of the email.

**Return-Path** – The actual address the email reply will be delivered to.

**Received:** The log of every server the email passed through before getting to the recipient.

**Authentication-Results:** This indicates if the email passed security checks or otherwise.

## Email Header Contents

**From:** Microsoft account team ,\_<no-reply@access-accsecurity.com>

**Reply-To:** [solutionteamrecognizd03@gmail.com](mailto:solutionteamrecognizd03@gmail.com)

**To:** phishing@pot

**Date:** Wed, 16 Aug 2023

**Subject:** Microsoft account unusual signin activity

**Return-Path:** [bounce@atujpdfghher.co.uk](mailto:bounce@atujpdfghher.co.uk)

**Signed-by:** null

**Received:** There are logs from three servers.

**Authentication-Results:** spf=none (sender IP is 89.144.44.41)

## Detected Anomalies

MX Lookup

mx:atujpdfghher.co.uk

Find Problems

Solve Email Delivery Problems

mx

| Test                     | Result                                                                                                               |                           |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------|
| DNS Record Published     | DNS Record not found                                                                                                 | <a href="#">More Info</a> |
| DMARC Record Published   | No DMARC Record found                                                                                                | <a href="#">More Info</a> |
| DMARC Policy Not Enabled | It is recommended to use a quarantine or reject policy. To enable BIMI, it is required to have one of these at 100%. | <a href="#">More Info</a> |

dns lookup

dns check

dmARC lookup

spf lookup

dns propagation

Reported by [nsb.nic.uk](#) on 10/8/2025 at 12:01:13 PM (UTC -5), [just for you.](#) [Transcript](#)

00.511.52.5

Relay (Seconds)

| Ho p | Delay     | From                                                                          | By                                                           | With                                                                                 | Time (UTC)            | Blackli st |
|------|-----------|-------------------------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------|------------|
| 1    | *         | MW2NAM04FT048.eop-NAM04.prod.protection.outlook.com 2603:10b6:303:85:cafe::78 | MW4PR04CA0179.outlook.office365.com 2603:10b6:303:85::34     | Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) | 8/16/2023 12:15:44 AM |            |
| 2    | 1 Second  | MW4PR04CA0179.namprd04.prod.outlook.com 2603:10b6:303:85::34                  | IA1PR19MB6449.namprd19.prod.outlook.com 2603:10b6:208:38b::5 | Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) | 8/16/2023 12:15:45 AM |            |
| 3    | 0 seconds | IA1PR19MB6449.namprd19.prod.outlook.com ::1                                   | MN0PR19MB6312.namprd19.prod.outlook.com                      | HTTPS                                                                                | 8/16/2023 12:15:45 AM |            |

[Gmail & Yahoo are now requiring DMARC - Get yours setup with Delivery Center](#)

### SPF and DKIM Information

#### Headers Found

| Header Name            | Header Value                         |
|------------------------|--------------------------------------|
| Authentication-Results | spf=none (sender IP is 89.144.44.41) |

- The reply-to field has a different content than the return-path. This suggests that the protocols have been tampered with and the receiver is being deceived.
- The message was not signed.
- The sender claims to be from **Microsoft Team** but the domains suggest differently.
- The recipient's mail servers tried to verify if the sender was authorized to send mail with the domain but no Sender Policy Framework (SPF) was found. This raises suspicions.
- No DNS and DMARC records were found.
- No DMARC policy was found.
- One of the received servers has been blacklisted.
- The sender IP has been reported 32 times on abuse IPDB.

## AbuseIPDB » 89.144.44.4

Check an IP Address, Domain Name, or Subnet  
e.g. 102.91.102.157, microsoft.com, or 5.188.10.0/24

**89.144.44.4** was found in our database!

This IP was reported **32** times. Confidence of Abuse is **0%**:

0%

|             |                                                                                           |
|-------------|-------------------------------------------------------------------------------------------|
| ISP         | GHOSTNET GmbH                                                                             |
| Usage Type  | Data Center/Web Hosting/Transit                                                           |
| ASN         | AS58212                                                                                   |
| Hostname(s) | 4.0-255.44.144.89.in-addr.arpa                                                            |
| Domain Name | ghostnet.de                                                                               |
| Country     |  Germany |

### Risk Assessment

The phishing attempt poses a high level of risk because one out of three servers has been blacklisted, no DNS and DMARC record and no DMARC policy was found. This shows invisibility from sender authentication, bypassing of anti-spoofing checks and the mimicking of a trusted entity, **Microsoft**.

### Recommendations

- Report email as spam.
- Block or isolate the sender and return-path email addresses.
- If there are attachments like links and files, they should not be directly interacted with.
- If the recipient is a technical person or a security analyst, the reputation of domains and IP should be verified, preventive measures should be implemented and, the analysis should be documented and reported.